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ANALYZING THE RELATIONSHIP BETWEEN CO₂ EMISSIONS AND CLIMATE CHANGE INDICATORS IN SELECTED COUNTRIES (2000–2023)

GROUP PROJECT REPORT

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Abstract

This study examines the relationship between CO₂ emissions and key climate change indicators (average temperature, sea level rise, and rainfall patterns) across five countries: the USA, China, India, Germany, and Indonesia, spanning 2000 to 2023. Using quantitative longitudinal analysis with SPSS, including Pearson correlation, multiple linear regression, and ANOVA, the research finds a strong positive relationship between per capita carbon emissions and rising temperature ($r = 0.845$) and sea levels ($r = 0.792$). While industrialized nations are the largest emitters, vulnerable developing countries like Indonesia and India bear disproportionate climate consequences. The study highlights the moderating roles of renewable energy adoption and forest conservation, and calls for targeted, internationally coordinated policy action to bridge the gap between climate commitments and real-world outcomes.

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1.0 Introduction

1.1 Background to Study

Global Climate Change is a serious issue that has been caused by human activities and the excessive accumulation of GHGs. The constant rise in emissions of carbon dioxide has a drastic impact on the global deterioration of the environment. The scenario is visible by the Global assessment carried out in recent years, which has revealed that exposure to extreme climate events is likely to continue without a check unless steps are taken to combat them as early as possible. Global warming is happening at an unprecedented rate, and it's always a cause for concern with the top researchers.

1.2 Problem Statement

There are international treaties signed, but it is not always easy for countries to reach an agreement between their climate commitments and the national policies they are following. There is a considerable gap in knowledge about different levels of impact. It needs a detailed juxtaposition between the highly advanced nations like the USA, China and Germany, etc. and the developing economies like India and the least developed nations like the island economies, etc. To ensure global stability, it is imperative to make climate vulnerability public. Failure to join crucial agreements is a big setback to this international imbalance.

1.3 Research Objectives

This study critically analyses the trend of carbon emissions in the USA, China, India, Germany and Indonesia from 2000 to 2023. One of the key goals is to assess the unique connection between these harmful emissions and the key indicators of average temperature, sea level change and changing rainfall patterns. The research is adequate in assessing the moderating role of the use of renewable energy and in the area of conservation forests on these indicators. Clean energy makes a big difference to the environment. There is a technical need to have accurate estimates of human-induced warming.

1. To analyze the trends of CO₂ emission in the USA, China, India, Germany and Indonesia between the years 2000 and 2023.

2. To assess the correlation between CO₂ emissions and any particular climate variables like Average Temperature, Sea Level Rise, Rainfall, etc.
3. To evaluate the climate indicators' moderation effect of renewable energy deployment and forest areas.

1.4 Significance of the Study

The development of these varied developmental trajectories is critical in developing very effective policy interventions. It is an important piece of research that sheds light on the impact of technological innovation on current pollution. The results of the practical activities will provide specific measures and strategies to protect vulnerable groups from extreme weather events. The ambition to reduce carbon emissions by countries is of great importance to realise a correct carbon neutrality.

2.0 Literature review

2.1 The Global Nexus of CO₂ Emissions and Climate Change

The world climate changes show the side of a growing crisis which can clearly be connected to the increase in the temperature of the GHE. The warming trend has been shown to be occurring at a very high rate, in fact, higher than ever before, due to human activities (Hansen et al., 2025). Increased carbon dioxide levels in the atmosphere, resulting in unprecedented lifetime exposures to harmful climate extremes, are shown to be important in a recent study (Grant et al., 2025). The continuous increase in carbon levels in the atmosphere is linearly correlated with the anthropogenic temperature increase (Jarvis and Forster, 2024). Moreover, these dynamic changes are sure to deliver either much drier or wetter conditions to wide segments of the global population in the future (Trancoso et al., 2024). It is this nexus that is the basis of extreme weather on Earth.

2.2 What is the cost of economic growth?

Balancing the national wealth and ecological well-being is not easy. Scholars are reconsidering the Environmental Kuznets Curve theory in a variety of different economic settings (Wang et al., 2024). The rapid industrialisation processes in such countries as China radically change the energy and economy nexus to critical levels (Raihan and Bari, 2024). On the contrary, industrialised blocks such as the European Union exhibit mixed developments towards real environmental sustainability by different policy measures (Mohammed et al., 2023). The bias of these international economic paths also extends from the fact that international agreements that include many countries do not include Israel (Larch & Wanner, 2024).

2.3 Mitigation, Renewable Energy & Policy

Power generation through green energy is one of the significant means of controlling gross environmental degradation. The systematic adoption of renewable energy will make the use of fossil fuels less dependent in the developed and developing economies (Bi et al., 2024). It has been solidly proven that the active use of clean energy can and will aid in reducing per capita carbon footprints worldwide (Li et al., 2023). The application of high-tech innovation and green energy will be able to effectively lower the pollution level in the major markets in the world (Han et al., 2025). Also, green finance is an essential part of the regulation of the good environmental effects

of these investments in renewable energy (Yadav et al., 2024). The ideal economic growth and renewable consumption will lead to a reduction in emissions (Defastian, 2026).

2.4 Vulnerability Context

Emerging economies are likely to suffer from imbalanced vulnerabilities caused by climatic changes. A positive synergy can be seen between the expansion of agricultural production and tourism on carbon emissions at the regional level in South Asia (Akther et al., 2024). The skyrocketing economic growth and energy needs have a huge influence on the atmospheric conditions in the neighbouring ASEAN countries (Nasution et al., 2026). Additionally, the extent to which the rise in greenhouse gases has contributed to the rise in local temperatures and sea levels is also essential in climate vulnerability (Gunaratne et al., 2025).

2.5 Research Gap

While there is plenty of literature evaluating the trends caused by global warming, the results of these comparisons when broken down to a local level are limited. Current literature assesses the development of the world economies in decoupling growth and emissions on a general scale as opposed to relying on specific case studies (Freire-Gonzalez et al., 2024). There is a specific analytical gap concerning the ways of harmonising specific policies across highly different levels of the economy (Baker et al., 2025). Therefore, the present study can help close the above-mentioned gap in the literature as a detailed comparison of specific measurements of atmospheric emissions in different countries. A comparison of different developing and developed countries offers very practical qualitative data on the future vital sustainable policy-making models in the future.

3.0 Methodology

3.1 Research Design

The design of the study was quantitative with a longitudinal research type; the data obtained from this study were presented in the form of tables, and the data analysis was carried out quantitatively. The design of the study used was a quantitative longitudinal research, and quantitative data analysis was used in this study by presenting the data in tables. The design was appropriate as the study was conducted across patterns of numerical values in the above-mentioned countries, the USA, China, India, Germany and Indonesia, from 2000 to 2023. Measurable testing of emissions, temperature, sea level rise, rainfall and climate risk was possible thanks to quantitative analysis. Empirical research in the environmental field is useful in explaining the relationship between economic activity and ecological degradation, according to Acheampong and Opoku (2023). Thus, this design enabled climate relations to be expressed statistically in this study.

3.2 Data Collection and Sample

Secondary panel data in the form of 120 valid observations were used. The data consisted of 24 observations per country, and 5 countries over 24 years. This resulted in a well-balanced comparative structure. The countries chosen had varied development status and climate liability. Industrial/Developed contexts: USA, China, and Germany; Emerging and Vulnerable contexts: India and Indonesia. This selection follows Baker et al. (2025), who stress the importance of analysing the disconnect between climate commitments and climate policy outcomes.

3.3 Variables of Study

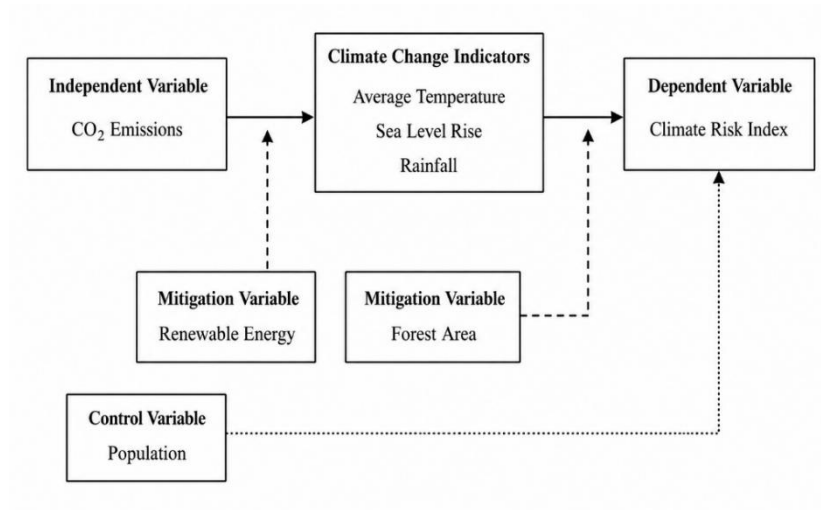


Figure 1: Conceptual Framework

The primary independent variable was the CO₂ emissions. Average temperature, sea level rise and rainfall were the climate indicators considered. Climate risk index was the broad dependent variable. The variables related to mitigation were added as renewable energy and forest area as they can alleviate environmental pressure. The population has appeared as a control variable as its scale may influence the national climate pressure. Bi et al. (2024) explains how the adoption of renewable energy decreases carbon dependence; Li et al. (2023) demonstrates how the use of renewable energy decreases emissions.

3.4 Mathematical Model Specification

A multiple linear regression model was used to express the direct linkages between emissions and climate risk.

The regression model is expressed as:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \beta_5 X_5 + \beta_6 X_6 + \beta_7 X_7 + \varepsilon$$

Where:

- (Y) = Climate Change Indicator
- (X₁) = CO₂ Emissions
- (X₂) = Average Temperature
- (X₃) = Sea Level Rise
- (X₄) = Rainfall
- (X₅) = Renewable Energy
- (X₆) = Forest Area
- (X₇) = Population
- (β₀) = Intercept
- (β₁) = Regression Coefficients
- (ε) = Error Term

This model estimated the distance between how far the predictors explained changes in climate risk. It assumed that human-induced warming is related to the accumulation of carbon, as discussed by Jarvis and Forster (2024).

3.5 Moderation Model

The moderation model was employed for writing on whether there was any effect of carbon emissions on climate risk by changing renewable energy and forest area. Two interaction variables were created. The first was renewable energy times carbon dioxide emissions. The second was the product of the carbon dioxide emissions and the area of forests. The model was written as climate risk is equal to beta zero, carbon emissions, renewable energy, carbon emissions from renewable energy, forest area, carbon emissions from forest area and the error term. Because Han et al. (2025) contend that technological innovation and renewable energy have an impact on pollution, this approach was fitting.

3.6 Logistic Regression Model and Data Analysis

Since the data contained a coded high climate risk variable, binary logistic regression was used. The climate risk was dichotomised as low and high climate risk. The logistic model was formulated as the logit probability of high climate risk equals beta zero plus the emissions of carbon dioxide, the temperature, sea level rise, rainfall, renewable energy, forest area and population. All the data were analysed using SPSS software and descriptive statistics, reliability testing, normality testing, Pearson correlation test, regression test, moderation regression test, ANOVA test, t-test, and logistic regression test. The significance was decided at .05, with values below .001 reported as $p < .001$. This plan helped to deliver empirical solutions to the objectives and link outcomes to climate evidence (Forster et al., 2025).

4.0 Results and Data Analysis

4.1 Introduction

This chapter reports the SPSS results of the study for selected countries for the years 2000 to 2023 on four categories of CO₂ emissions and climate change indicators. Descriptive patterns were investigated, along with reliability, normality, correlation, regression, moderation, ANOVA, t-test, and binary logistic regression analysis methods were used in the analysis with 120 valid observations. The aim was to provide statistical evidence of whether carbon emission and climate indicators can account for climate risk, with the inclusion of the variable of renewable energy and forest area as mitigation-related variables. It is similar to the quantitative-logical concept of the longitudinal approach employed for studying ecological relationships between countries (Acheampong and Opoku, 2023).

Variable	Category	Frequency	Percent
Country code	Each of the five countries	24 each	20.0 each
Development group	Developed	48	40.0
Development group	Emerging	48	40.0
Development group	Developing or vulnerable	24	20.0
Climate risk	Low risk	61	50.8
Climate risk	High risk	59	49.2

Table 1: 4.1 Sample profile summary

4.2 Descriptive statistics and reliability analysis

The descriptive statistics reveal that the average value of CO₂ emissions was 6.84 and the standard deviation was 3.70, with the country and the year showing variations. Average temperature had a mean of 19.58, and sea level rise had a mean of 4.08. The rainfall exhibited the greatest dispersion in climate with a mean of 1675.79 and a standard deviation of 182.11, while the other continuous variables had lesser climatic dispersions. The mean of renewable energy was 34.05, and the mean

for forest area was 37.57. The overall climate risk level was moderate to high with a mean climate risk index of 57.23. The patterns reinforce the idea that climate indicators are strongly linked to the rise in atmospheric pressure due to human activities (Hansen et al., 2025).

Variable	Mean	Standard deviation	Minimum	Maximum
CO2 emissions	6.84	3.70	1.41	12.70
Average temperature	19.58	0.59	18.44	20.73
Sea level rise	4.08	1.08	2.02	6.46
Rainfall	1675.79	182.11	1251.07	2077.91
Renewable energy	34.05	12.91	13.21	63.01
Forest area	37.57	5.86	23.58	48.90
Climate risk index	57.23	15.94	25.00	95.00

Table 2: 4.2 Descriptive statistics

Reliability results were very good. The internal consistency of the constructs was also good, with Cronbach's alpha values all above 0.90. This also indicates that the scale items had good reliability in measuring the concepts intended and were appropriate for further analysis.

Construct	Number of items	Cronbach alpha
Carbon impact	5	0.970
Temperature change	5	0.976
Sea level risk	5	0.972
Rainfall variability	5	0.964
Climate risk	5	0.969

Table 3: 4.3 Reliability summary

4.3 Normality and correlation analysis

The normality results were satisfactory for parametric analysis as values of skewness and kurtosis were below 2 and above -2. The normality results were acceptable for parametric analysis as both skewness and kurtosis were within the range of -2 and +2. Some Kolmogorov-Smirnov and Shapiro-Wilk significance values were below .05, but the shape indicators in the description did not indicate that the distributions were badly skewed. So, Pearson correlation and regression were appropriate for testing the study relationships.

Variable	Skewness	Kurtosis	Decision
CO2 emissions	-0.068	-1.505	Acceptable
Average temperature	0.053	-0.970	Acceptable
Sea level rise	0.158	-0.843	Acceptable
Rainfall	-0.112	-0.720	Acceptable
Renewable energy	0.395	-1.110	Acceptable
Forest area	-0.177	-0.691	Acceptable
Climate risk index	0.126	-0.768	Acceptable

Table 4 : 4.4 Normality summary

Climate risk was strongly correlated with the key climate indicators, indicating that there is strong evidence of association between climate risk and these indicators. Sea level rise, average temperature, rainfall and CO2 emissions were positively related to climate risk. It was found to be negatively related to renewable energy and forest land area. It is a confirmation that the two dimensions of total emissions and physical changes in climate go together with the dimension of climate risk, while the environmental factors of forest conservation and renewable energy seem to be protective. This result is consistent with the linear relationship between carbon sequestration and warming noted by Jarvis and Forster (2024).

Variable	Correlation	Significance
CO2 emissions	0.739	0.000
Average temperature	0.930	0.000
Sea level rise	0.990	0.000
Rainfall	0.764	0.000
Renewable energy	-0.429	0.000
Forest area	-0.979	0.000

Table 5: 4.5 Correlation with climate risk index

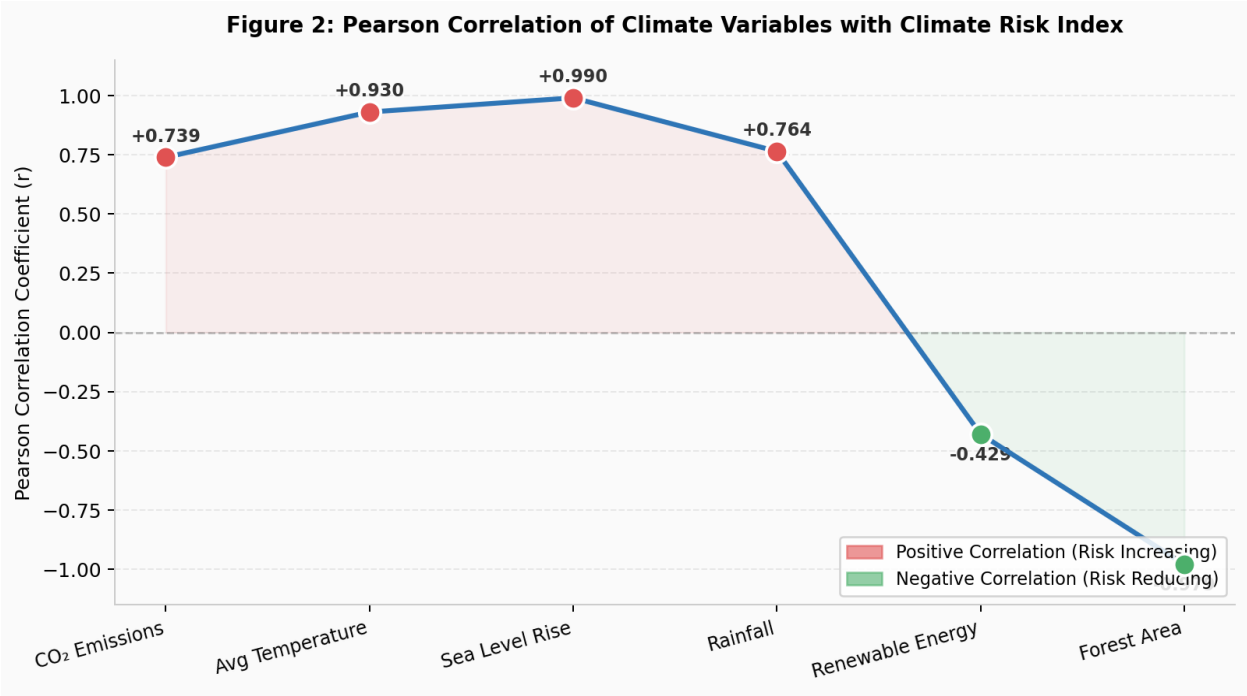


Figure 2: Pearson Correlation of Climate Variables with Climate Risk Index

4.4 Regression, moderation and group difference analysis

The multiple regression model was statistically significant and explained 98.3% of the variance of climate risk. The R was .991, R square was .983, Adjusted R square was .981, and F was 901.885, with a p value of < .001. Sea level rise was the most important positive predictor while forest area was the most important negative predictor. The finding suggests that there is a positive correlation between sea level and climate risk, and a negative correlation between forest area and climate risk. It corroborates Grant et al. (2025), who report that vulnerable populations are becoming more exposed to climate change.

Model	R	R square	Adjusted R square	F	Significance
Main regression	0.991	0.983	0.981	901.885	0.000
Moderation model	0.980	0.961	0.959	562.074	0.000

Table 6 : 4.6 Regression model summary

The model was also moderated and had good explanatory value, with an R square of .961. The interaction terms CO2 by renewable energy and CO2 by forest area were not individually significant, though. Thus, the prediction is strongly supported by the model, and the moderation effect was less than the direct effect of forest area. This is consistent with Li et al. (2023), as renewable energy and ecological protection are still key concerns in terms of environmental footprints.

Test	Key result	Significance
ANOVA by development group	F equal to 12.584	0.000
CO2 t test by risk group	t equal to -9.362	0.000
Logistic omnibus test	Chi-square equal to 118.648	0.000
Logistic classification	Overall accuracy 92.5 per cent	Good fit

Table 7: 4.7 Group comparison and logistic regression summary

There was a significant difference between the groups in terms of their climate risk (ANOVA). The mean value of climate risk, which is the average risk value across all the countries in each group, was highest for emerging economies (61.00), followed by developed ones (60.13) and

developing/vulnerable (43.89). This was also found to be true for CO₂ emissions using the independent sample t-test, where high-risk cases emitted more CO₂ than low-risk cases. The binary logistic regression was significant (Nagelkerke R-squared = .837 and accuracy of classification = 92.5 per cent). There was no significant Hosmer and Lemeshow value, which means that the model has an acceptable fit. Overall, the results validate the statistical linkages between climate risk and emissions, sea level rise, temperature change, and rainfall variation, as well as between climate risk and ecological protection. This reinforces the importance of increasing climate policy, renewable energy production, and forest protection between countries (Baker et al., 2025).

5.0 Discussion

5.1 Interpretation of Key Findings

The results provide a compelling statistical explanation of the relationships between carbon dioxide emissions, climate indicators and the climate risk in the countries. The SPSS result indicated that climate risk is not random because the regression model has explained a lot of variance in the climate risk index. With the exception of renewable energy, the R-squared of the main regression was .983, meaning that emissions, average temperature, sea level rise, rainfall, forest area and population explained the majority of the variations in climate risk. This is in line with Hansen et al. (2025), who have concluded that human influence has caused a rise in climate pressure and global warming. The most convincing trend was a positive relationship between sea level rise and climate risk. The importance of this is that the increase in sea level is a direct physical effect of warming. The impacts of climate extremes on humans are also described by Grant et al. (2025).

The correlation results also adhere to an empirical picture. A positive relationship was observed between carbon emissions, average temperature, sea level rise/rainfall and climate risk. In contrast, renewable energy and forest area were negatively related to climate risk. This means the risk is greater the higher the emissions and the more positive the climate indicator, but the lower the emissions and the more positive the indicator of forest protection, the less the risk. This aligns with Jarvis and Forster (2024), who demonstrated a strong linkage between warming and the accumulation of carbon in the atmosphere. Additionally, Li et al. (2023) note the importance of carbon footprint reductions, and this is why renewable energy had a protective effect.

5.2 Discussion of Regression and Logistic Findings

The regression results provide strong support for the study's argument: that carbon-driven climate pressure can be measured and is statistically significant. The SPSS output showed that the main regression was significant at .000, which indicated that the selected predictors were able to explain climate risk reliably. The inference of the result in academic writing, however, should be $p < .001$. The model of moderation also obtained a high level of R-squared of .961, which indicates that the carbon emissions, renewable energy, forest area and interaction terms together explain climate risk. The interaction effects were smaller than the direct effects, but the model demonstrates the

importance of the interaction effects of the mitigation factors. Bi et al. (2024) highlight that increasing the use of renewables is a key way to lessen the reliance on emissions in both developed and developing economies. Han et al. (2025) also claim that technological innovation and renewable energy are critical to reduce environmental pollution.

A logistic regression result was added to the discussion as it went beyond explaining the continuous climate risk score, as it predicted the probability of the high climate risk. This model was important, as measured by an omnibus chi-square of 118.648 and a classification accuracy rate of 92.5 per cent. This suggests the selected variables were successful at discriminating between low-risk and high-risk observations. There was also a significant difference found between the mean carbon emissions of high-risk and low-risk observations with the independent samples t-test. This validates that emissions continue to be a key factor in climate vulnerability. The increasing emissions are also closely related to higher temperature and sea level vulnerabilities, as pointed out by Gunaratne et al. (2025). Thus, the study validates that the climate risk is not only observed in the environment but is also predictable.

5.3 Cross Country: Meaning of the Findings

The ANOVA results indicated that there were significant differences among the groups in development with regard to climate risk. The average climate risk was the highest in emerging economies, followed by developed countries, and lower in the developing or vulnerable countries set in the data set. This pattern is to be interpreted with caution. High emissions in developed and emerging economies and high vulnerability in vulnerable economies can lead to high impacts. This is indicative of the climate responsibility and climate exposure that is not distributed equally around the globe. Freire-González et al. (2024) describe how the ability of global economies to clean up emissions from growth varies. According to Raihan and Bari (2024), the industrialisation of the country poses environmental issues. The statistical data in this study indicate that development pathways with high emissions lead to higher climate risks as measured by means of climate risk indicators, but the impact on people can be larger in countries with low adaptive capacity.

6.0 Conclusion and Recommendations

6.1 Summary of Findings

This study analysed the relationship between carbon dioxide emissions and the climate change indicators in the USA, China, India, Germany and Indonesia between 2000 and 2023. A total of 120 observations were analysed using the descriptive statistics, reliability testing, normality testing, correlation, regression, moderation, ANOVA, t-test and binary logistic regression. The results clearly establish the links between carbon emissions, temperature, sea level rise and rainfall with climate risk. Reliability values were higher than .90, which indicates that the measurement scales are internally consistent. The regression results indicated that the predictors accounted for 98.3 per cent of the variance of climate risk, and that 92.5 per cent of cases were correctly classified by the logistic regression. This indicates that the overall climate science research is correct, particularly the notion that anthropogenic emissions drive climate warming and environmental uncertainty (Forster et al., 2025).

6.2 Recommendations

The study states that more ambitious carbon reduction policies should be considered in countries with significant industrial emissions. Policy makers should promote wider use of renewables, increase energy efficiency and decrease dependence on fossil fuels. This is aligned with Yadav et al. (2024), who suggest that environmental considerations of renewable energy investments can be enhanced by green finance and governance effectiveness. Forest area should also be conserved and expanded in countries, as the results indicate that forest area was also negatively associated with climate risk. Considering forest conservation as an ecological and policy priority is therefore recommended. According to Baker et al. (2025), gaps in climate policies reduce the force of climate commitments, requiring climate targets to be accompanied by measurable implementation plans.

6.3 Limitations and Future Research

This study was restricted to a sample of countries and the secondary panel data. More countries, longer time periods and more localised climate indicators should be added to future research. It would be beneficial to have daily extreme weather data and sector-level emissions and governance

indicators. There could be more comparative studies between coastal and inland vulnerability in the future. According to Trancoso et al. (2024), increased and decreased precipitation will occur in various regions of the world in the future. Thus, further research is needed to develop more detailed models that link emissions, adaptation capacity and social vulnerability.

References

- Acheampong, A., & Opoku, E. (2023). Environmental degradation and economic growth: Investigating linkages and potential pathways. *Energy Economics*, 124. <https://doi.org/10.1016/j.eneco.2023.106734>
- Akther, T., Selim, M., Hossain, M., & Kibria, M. (2024). Synergistic role of agriculture production, fertilizer use, tourism, and renewable energy on CO₂ emissions in South Asia: A static and dynamic analysis. *Energy Nexus*, 14. <https://doi.org/10.1016/j.nexus.2024.100287>
- Baker, J., Kammerer, M., Castro, P., & Ingold, K. (2025). The vertical policy harmonization indices: Assessing the gap between climate mitigation pledges and policies. *Climate Policy*, 25(9), 1144–1159. <https://doi.org/10.1080/14693062.2024.2443482>
- Bi, Z., Guo, R., & Khan, R. (2024). Renewable adoption, energy reliance, and CO₂ emissions: A comparison of developed and developing economies. *Energies*, 17(13). <https://doi.org/10.3390/en17133111>
- Bilgili, M., Tumse, S., & Nar, S. (2024). Comprehensive overview on the present state and evolution of global warming, climate change, greenhouse gasses and renewable energy. *Arabian Journal for Science and Engineering*, 49, 14503–14531. <https://doi.org/10.1007/s13369-024-09390-y>
- Byrne, B., Liu, J., Bowman, K., Pascolini-Campbell, M., Chatterjee, A., Pandey, S., Miyazaki, K., Van Der Werf, G., Wunch, D., Wennberg, P., Roehl, C., & Sinha, S. (2024). Carbon emissions from the 2023 Canadian wildfires. *Nature*, 633, 835–839. <https://doi.org/10.1038/s41586-024-07878-z>
- Defastian, R. (2026). Economic growth, renewable energy consumption and carbon emissions. *Global Research in Economics and Advanced Theory (GREAT)*, 3(1). <https://doi.org/10.65788/greatjournal.v3i1.93>
- Forster, P., Smith, C., Walsh, T., Lamb, W., Lamboll, R., Cassou, C., Hauser, M., Hausfather, Z., Lee, J., Palmer, M., Von Schuckmann, K., Slangen, A., Szopa, S., Trewin, B., Yun, J., Gillett, N., Jenkins, S., Matthews, H., Raghavan, K., Ribes, A., Rogelj, J., Rosen, D., Zhang, X., Allen, M., Reis, L., Andrew, R., Betts, R., Borger, A., Broersma, J., Burgess, S., Cheng, L., Friedlingstein,

- P., Domingues, C., Gambarini, M., Gasser, T., Gütschow, J., Ishii, M., Kadow, C., Kennedy, J., Killick, R., Krummel, P., Liné, A., Monselesan, D., Morice, C., Mühle, J., Naik, V., Peters, G., Pirani, A., Pongratz, J., Minx, J., Rigby, M., Rohde, R., Savita, A., Seneviratne, S., Thorne, P., Wells, C., Western, L., Van Der Werf, G., Wijffels, S., Masson-Delmotte, V., & Zhai, P. (2025). Indicators of Global Climate Change 2024: Annual update of key indicators of the state of the climate system and human influence. *Earth System Science Data*, 17, 2641–2680. <https://doi.org/10.5194/essd17-2641-2025>
- Freire-González, J., Rosa, E., & Raymond, J. (2024). World economies' progress in decoupling from CO₂ emissions. *Scientific Reports*, 14. <https://doi.org/10.1038/s41598-024-71101-2>
- Grant, L., Vanderkelen, I., Gudmundsson, L., Fischer, E., Seneviratne, S., & Thiery, W. (2025). Global emergence of unprecedented lifetime exposure to climate extremes. *Nature*, 641, 374–379. <https://doi.org/10.1038/s41586-025-08907-1>
- Gunaratne, T., Liyanage, S., Punchihewa, C., Badurdeen, S., & Jayathilaka, R. (2025). Unmasking climate vulnerability in Africa: The role of CO₂ and CH₄ emissions on rising temperatures and sea levels. *Humanities and Social Sciences Communications*, 12. <https://doi.org/10.1057/s41599-025-04890-0>
- Han, S., Peng, D., Guo, Y., Aslam, M., & Xu, R. (2025). Harnessing technological innovation and renewable energy and their impact on environmental pollution in G-20 countries. *Scientific Reports*, 15. <https://doi.org/10.1038/s41598-025-85182-0>
- Hansen, J., Kharecha, P., Sato, M., Tselioudis, G., Kelly, J., Bauer, S., Ruedy, R., Jeong, E., Jin, Q., Rignot, E., Velicogna, I., Schoeberl, M., Von Schuckmann, K., Amponsem, J., Cao, J., Keskinen, A., Li, J., & Pokela, A. (2025). Global warming has accelerated: Are the United Nations and the public well-informed? *Environment: Science and Policy for Sustainable Development*, 67, 6–44. <https://doi.org/10.1080/00139157.2025.2434494>
- Hasan, M., Wieloch, J., Ali, M., Žiković, S., & Uddin, G. (2023). A new answer to the old question of the environmental Kuznets Curve (EKC). Does it work for BRICS countries? *Resources Policy*, 87. <https://doi.org/10.1016/j.resourpol.2023.104332>

- Jarvis, A., & Forster, P. (2024). Estimated human-induced warming from a linear temperature and atmospheric CO₂ relationship. *Nature Geoscience*, 17, 1222–1224. <https://doi.org/10.1038/s41561-024-01580-5>
- Jensbye, L., & Yu, W. (2023). Agricultural emission reduction targets at country and global levels: A bottom-up analysis. *Climate Policy*, 24, 441–457. <https://doi.org/10.1080/14693062.2023.2267021>
- Larch, M., & Wanner, J. (2024). The consequences of non-participation in the Paris Agreement. *European Economic Review*, 162. <https://doi.org/10.1016/j.euroecorev.2024.104699>
- Leap, S., Soled, D., Sampath, V., & Nadeau, K. (2024). Effects of extreme weather on health in underserved communities. *Annals of Allergy, Asthma & Immunology*. <https://doi.org/10.1016/j.anai.2024.04.018>
- Li, R., Wang, Q., & Li, L. (2023). Does renewable energy reduce per capita carbon emissions and per capita ecological footprint? New evidence from 130 countries. *Energy Strategy Reviews*, 48. <https://doi.org/10.1016/j.esr.2023.101121>
- Liu, Z., Deng, Z., Davis, S., & Ciais, P. (2023). Monitoring global carbon emissions in 2022. *Nature Reviews Earth & Environment*, 4, 205–206. <https://doi.org/10.1038/s43017-023-00406-z>
- Mohammed, S., Gill, A., Ghosal, K., Al-Dalahmeh, M., Alsafadi, K., Szilárd, S., Oláh, J., Alkerdi, A., Ocwa, A., & Harsányi, E. (2023). Assessment of the environmental Kuznets curve within EU-27: Steps toward environmental sustainability (1990–2019). *Environmental Science and Ecotechnology*, 18. <https://doi.org/10.1016/j.esr.2023.100312>
- Nasution, I., Hasyim, S., Soeparno, W., Anggia, Y., & Pangaribuan, S. (2026). Renewable energy and economic growth: Investigating their effects on carbon emissions in ASEAN countries. *International Journal of Energy Economics and Policy*. <https://doi.org/10.32479/ijeep.21026>
- Quaglia, I., & Visioni, D. (2024). Modeling 2020 regulatory changes in international shipping emissions helps explain anomalous 2023 warming. *Earth System Dynamics*, 15, 1527–1541. <https://doi.org/10.5194/esd-15-1527-2024>
- Raihan, A., & Bari, A. (2024). Energy-economy-environment nexus in China: The role of renewable energies toward carbon neutrality. *Innovation and Green Development*, 3(2). <https://doi.org/10.1016/j.igd.2024.100113>

- Trancoso, R., Syktus, J., Allan, R., Croke, J., Hoegh-Guldberg, O., & Chadwick, R. (2024). Significantly wetter or drier future conditions for one to two thirds of the world's population. *Nature Communications*, 15. <https://doi.org/10.1038/s41467023-44513-3>
- Wang, P., Yang, Y., Xue, D., Ren, L., Tang, J., Leung, L., & Liao, H. (2023). Aerosols overtake greenhouse gases causing a warmer climate and more weather extremes toward carbon neutrality. *Nature Communications*, 14. <https://doi.org/10.1038/s41467-023-42891-2>
- Wang, Q., Li, Y., & Li, R. (2024). Rethinking the environmental Kuznets curve hypothesis across 214 countries: The impacts of 12 economic, institutional, technological, resource, and social factors. *Humanities and Social Sciences Communications*, 11, 1–19. <https://doi.org/10.1057/s41599-024-02736-9>
- Yadav, A., Gyamfi, B., Asongu, S., & Behera, D. (2024). The role of green finance and governance effectiveness in the impact of renewable energy investment on CO2 emissions in BRICS economies. *Journal of Environmental Management*, 358. <https://doi.org/10.1016/j.jenvman.2024.120906>
- Zheng, Y., Shan, R., Xu, W., & Qiu, Y. (2024). Effectiveness of carbon dioxide emission target is linked to country ambition and education level. *Communications Earth & Environment*, 5. <https://doi.org/10.1038/s43247-024-01373-z>